

Name: Answersmax: 23 points

This task will look at transformation of the sine curve. "Transformation" in a maths context means we move around and resize the graph of a function. By applying transformations to the sine curve $y = \sin x$, we can generate the graph for the general sine function $y = a \sin(b(x-c)) + d$, where the parameters a , b , c , and d determine various features on the graph. Investigating the effects of a , b , c , and d , it is purposeful to split up the general sine curve into different function families.

Investigation 1 - the function family $y = a \sin x$

(a) Using GeoGebra, graph the function $y = f(x) = a \cdot \sin x$ and observe the changes various values of a apply to the graph of the function. Write down your observations using the appropriate mathematical terms (principal axis, amplitude, period).

- amplitude changes; the larger $|a|$, the larger the amplitude
- principal axis $y=0$ stays the same
- period stays the same

(b) Using your GDC, complete the following table.

a	Function	Maximum	Minimum	Period	Amplitude
1	$y = \sin x$	1	-1	2π	1
2	$y = 2 \sin x$	2	-2	2π	2
0.5	$y = 0.5 \sin x$	0.5	-0.5	2π	0.5
-1	$y = -\sin x$	1	-1	2π	1
a	$y = a \sin x$	$ a $	$- a $	2π	$ a $

Investigation 2 - the function family $y = \sin(bx)$ ($b > 0$)

(a) Using GeoGebra, graph the function $y = f(x) = \sin(bx)$ and observe the changes various values of b apply to the graph of the function ($b > 0$). Write down your observations using the appropriate mathematical terms (principal axis, amplitude, period, ...).

- period changes; the larger the value of " b ", the shorter the period
- principal axis stays $y=0$; amplitude stays 1

$$\frac{1}{3}$$

①

3

(c) $\frac{2\pi}{3}$ (2.09) (iv) 6π (18.8) (iii) $\frac{5\pi}{3}$ (5.24)

②

- Changing "c" moves the graph left or right (parallel to x-axis)

- if c is positive, the graph moves " c " units to the right
- if c is negative, the graph moves " $|c|$ " units to the left

 $\frac{1}{2}$

c	Function	Maximum	Minimum	Period	Amplitude
0	$y = \sin x$	1	-1	2π	1
-2	$y = \sin(x - 2)$	1	-1	2π	1
2	$y = \sin(x + 2)$	1	-1	2π	1
$-\frac{\pi}{3}$	$y = \sin(x - \frac{\pi}{3})$	1	-1	2π	1
c	$y = \sin(x - c)$	1	-1	2π	1

$$\sqrt{2}$$

"d" affects the vertical shift; changing "d" moves the graph up or down. If $d > 0$, it moves up about d units
 $d < 0$ down